Face Mask Detection using Transfer Learning with MobileNetV2

End-to-End Pipeline Flowchart

1 INPUT

- Import Libraries
- Set Training Parameters – Learning Rate, Epochs, Batch Size
- Load Image Paths from Dataset – Images of faces with and without masks



2 DATA PREPROCESSING

224×224 ×3

1. Initialize Empty Lists `data = []` `labels = []`
2. **FOR EACH IMAGE IN DATASET**
   a. Extract Class Label from Folder Name
   b. Load and Resize Image to the Input Size Expected by MobileNetV2 (224 × 224)
   c. Convert Image into NumPy Array
   d. Image becomes pixel numbers with shape 224 × 224 × 3
   e. Preprocess Input for Pretrained MobileNetV2
   f. Append Processed Image → data[]
   g. Append Label → labels[]
3. h. Convert data & labels into NumPy Arrays
4. i. **Label Binarization** – Convert class labels into binary numbers, e.g, without_mask → 0, with_mask → 1
5. j. **One-Hot Encoding** – Convert binary labels into categorical vectors:
   i. 0 → [1, 0]
   ii. 1 → [0, 1]
6. k. **Train/Test Split** – 80% training / 20% testing, stratified
7. l. **Data Augmentation** – Generate varied versions of training images using rotation, zoom, shifting, shearing, and flipping to improve generalization and reduce overfitting.

3 BUILD THE MODEL

MobileNetV2

Pretrained Feature Extractor

Classification Head

- **Load Pretrained MobileNetV2 for Feature Extraction** – Use ImageNet pretrained weights and remove the original classification layers (include_top=False).
- **Freeze Base Model**– Freeze MobileNetV2 layers so that their pretrained weights are not updated during the initial training.
- **Add Classification Head** – Add Average Pooling, Flatten, Dense, Dropout, and a final Dense output layer with Softmax activation on top of the pretrained MobileNetV2 to classify the extracted features into two classes: Mask and No Mask.

4 COMPILE THE MODEL

- **Binary Cross-Entropy Loss Function** – Measures the error between actual and predicted class values.
- **Adam Optimizer** – Uses the calculated loss to adjust the model's trainable weights and improve predictions.
- **Accuracy Metric** – Measures the percentage of correct predictions.

5 TRAIN AND VALIDATE THE MODEL

- Train the model using augmented training data and evaluate its performance on validation data after each epoch using loss and accuracy.

6 PREDICT TEST DATA

Mask 0.92
No Mask 0.08

- Pass unseen test images through the trained model to obtain class probabilities and predict whether each face has a Mask or No Mask.
- Then: **argmax** → Select the class with the highest predicted probability

7 EVALUATE THE MODEL

- Evaluate on Test Set using Classification **Report** – Analyze Precision, Recall, F1-score, and Support, along with Accuracy.

8 CHECK PERFORMANCE

Check Model Performance – Is the performance statisfactory?

YES → Save Model

NO → Tune Hyperparameters / Fine-tune Model → Retrain

Possible tuning:

- Learning rate
- Number of epochs
- Batch size
- Data augmentation
- Unfreeze selected MobileNetV2 layers for fine-tuning

A step towards a Safer & Healthier World!